# Preliminary Results

## Large Scale Optimization

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# 1. Convexity. Basic concepts

## Convex sets and convex functions

### Definition of a convex set $C$

A set $C \subseteq \mathbb{R}^n$ is convex if, given $x^1, x^2 \in C$, the closed segment $\overline{x^1, x^2}$ is contained in $C$, i.e.: $\overline{x^1, x^2} \subseteq C$,

$$\left( \overline{x^1, x^2} \stackrel{\Delta}{=} \{x \in \mathbb{R}^n \,|\, x = \alpha x^1 + (1-\alpha)x^2,\ 0 \leq \alpha \leq 1\} \right)$$

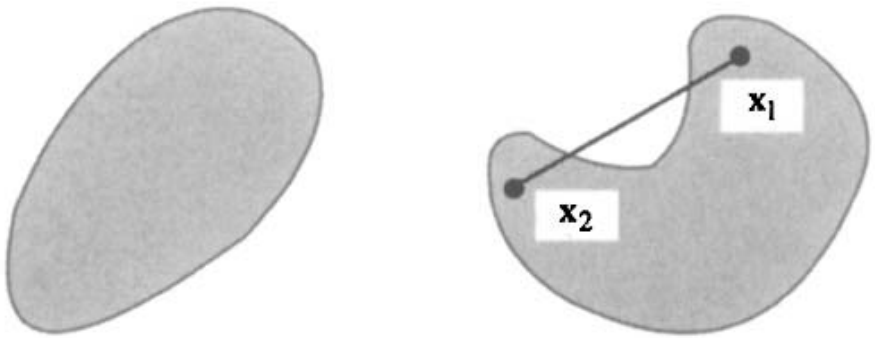



Figure: A convex set (left). A non-convex set (right)

If $C_1, C_2$ are convex sets, then $C_1 \cap C_2$, $C_1 \oplus C_2$, $C_1 \ominus C_2$ are also convex sets.

## Convex sets and convex functions

### Definition of a convex function $f$ on $P$, convex set

$$f(\alpha x^1+(1-\alpha)x^2) \leq \alpha f(x^1)+(1-\alpha)f(x^2),\ 0 \leq \alpha \leq 1,\ x^1, x^2 \in P$$

If $f$ is convex on $\mathbb{R}^n$: $\quad f(x) \geq f(x_1) + (x - x_1)^\top \nabla f(x_1)$.

Also, if $f$ is convex on $P$ (a convex set), then:

$f(x) \geq f(x_1) + (x - x_1)^\top \nabla f(x_1),\ \forall\, x, x_1 \in P$

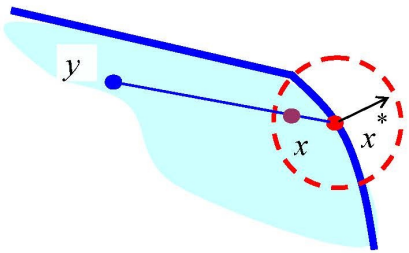
If $x^*$ is a local minimum of $f$ (convex) on $P$ (a convex set) $\Rightarrow$

$\Rightarrow x^*$ is a global minimum of $f$ over $P$

$\nabla f(x^*)^\top (x - x^*) \geq 0,\ \forall x \in B_\delta(x^*) \cap P$

If $y \in P,\ y \notin B_\delta(x^*) \cap P\ y\bar{x}^* \subset P$

$f(y) - f(x^*) \geq \nabla f(x^*)^\top (y - x^*) = \alpha \nabla f(x^*)^\top (x - x^*) \geq 0$



## Polyhedral Sets and extreme points

### Relevant examples of convex sets

- **Hyperplane** $S = \{x \in \mathbb{R}^n | \ a^\top x = \alpha\}$, where $a \neq 0$ and $\alpha \in \mathbb{R}$
- **Half-Space** $S = \{x \in \mathbb{R}^n | \ a^\top x \leq \alpha\}$, where $a \neq 0$ and $\alpha \in \mathbb{R}$
- **Polyhedral Set** $S = \{x \in \mathbb{R}^n | \ Ax \leq b\}$, where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Finite intersection of $m$ half-spaces.
- **Polyhedral Cone** $C = \{x \in \mathbb{R}^n | \ Ax \leq 0\}$, where $A \in \mathbb{R}^{m \times n}$. Finite intersection of $m$ half-spaces, each of them containing $0 \in \mathbb{R}^m$.

## Convex Hulls

**Convex Hull of a finite set of points $S = \{x^1, ..., x^k\}$ (Polytope)**

The convex hull of $S$, denoted by *Hull*$\{S\}$ (also as *Conv*$\{S\}$), is defined as:

$$\textit{Hull}\{S\} \triangleq \left\{ x \in \mathbb{R}^n \mid x = \sum_{\ell=1}^{k} \lambda_\ell x^\ell,\ \sum_{\ell=1}^{k} \lambda_\ell = 1,\ \lambda_\ell \geq 0,\ 1 \leq \ell \leq k \right\}$$

***Hull*$\{S\}$ for a general subset $S \subset \mathbb{R}^n$.**

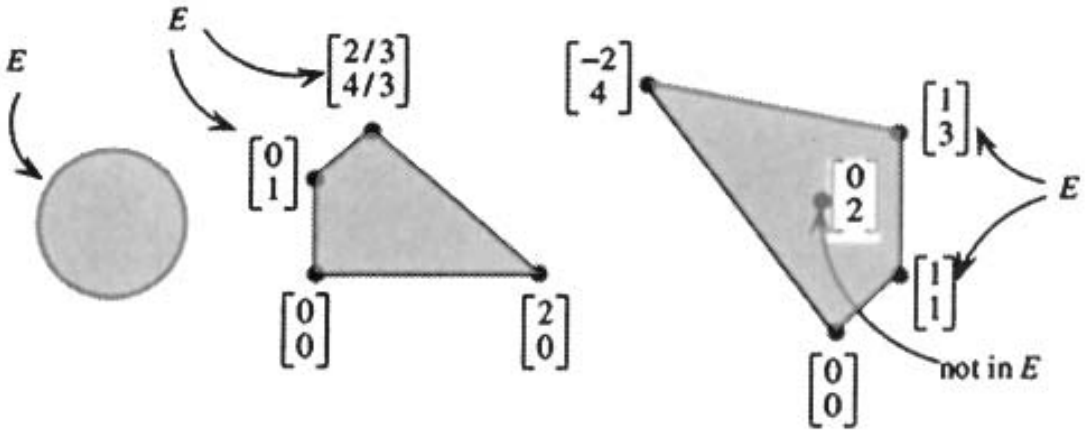
$$x \in \textit{Hull}\{S\} \Leftrightarrow \begin{cases} \exists \{x^1, ..., x^k\} \subseteq S, k > 0, \ \textit{and}\ \lambda_1, ...\lambda_k \geq 0, \\ \textit{such that} \displaystyle\sum_{\ell=1}^{k} \lambda_\ell = 1 \\ x = \displaystyle\sum_{\ell=1}^{k} \lambda_\ell x^\ell, \end{cases}$$

## Convex Hulls

### Extreme points (vertexes) of a Convex Set

$\hat{x} \in S$ is a vertex of $S$ iff. for any two points $x^1 \neq x^2 \in S$ so that $\hat{x} = \alpha x^1 + (1-\alpha)x^2$, $0 \leq \alpha \leq 1$ either $\hat{x} = x^1$ or $\hat{x} = x^2$.

$\{x \in \mathbb{R}^2 \mid x_1^2 + x_2^2 \leq 1\}$,
$\{x \in \mathbb{R}^2 \mid x_1 + x_2 \leq 2, -x_1 + 2x_2 \leq 2, x \geq 0\}$,
$Hull\{(0,0),(1,1),(1,3),(-2,4),(0,2)\}$



# 2. Useful concepts from the Simplex Algorithm

## Standard form and bfs

### STANDARD FORM OF AN L.P.

After suitable transformations any L.P. can be expressed as:

$$
\begin{aligned}
Min_x \quad & c_1 \cdot x_1 + \ldots + c_n \cdot x_n \\
s.a: \quad & a_{11} \cdot x_1 + \ldots + a_{1n} \cdot x_n = b_1 \\
& a_{21} \cdot x_1 + \ldots + a_{2n} \cdot x_n = b_2 \\
& \ldots \\
& a_{m1} \cdot x_1 + \ldots + a_{mn} \cdot x_n = b_m \\
& x_1 \geq 0, \ \ldots x_n \geq 0
\end{aligned}
\quad \longrightarrow \quad
\begin{aligned}
Min_x \quad & c^\top x \\
s.a: \quad & Ax = b \\
& x \geq 0
\end{aligned}
$$

$(m \leq n)$

- All the variables $x_i$ are subject to $x_i \geq 0$, $i = 1, 2, \ldots n$
- Any rhs term $b_i$ are non-negative: $b_i \geq 0$, $i = 1, 2, \ldots m$
- Matrix $A$ is full rank:

Therea are m columns of $A$ such that when building a squared matrix $B$ with them, $B$ is inversible.

Any LP solver converts automatically to the standard form.

## Standard form and bfs

**Example:**

$$\begin{array}{lll} Min & -x_1 + 3x_2 + 4x_3 & \\ s.a: & -x_1 - x_2 - x_3 & \geq -5 \\ & 2x_1 + 3x_2 + x_3 & \geq 6 \\ & x_2, \ x_3 \geq 0 & \end{array}$$

$$\begin{array}{lll} Min & -x_1 + 3x_2 + 4x_3 & \\ s.a: & -x_1 - x_2 - x_3 - x_4 & = -5 \\ & 2x_1 + 3x_2 + x_3 - x_5 & = 6 \\ & x_2, \ x_3, \ x_4, \ x_5 \geq 0 & \end{array}$$

$$x_1 = 5 - 2x_2 - x_3 - x_4$$

$$c^\top x = -5 + 2x_2 - x_3 - x_4 + 3x_2 + 4x_3 = \cancel{-5} + 5x_2 + 5x_3 + x_4$$

$$2(5 - x_2 - x_3 - x_4) + 3x_2 + x_3 - x_5 = 6$$

$$\begin{array}{lll} Min & 5x_2 + 5x_3 + x_4 & \\ s.a: & x_2 + x_3 + 2x_4 + x_5 & = 4 \\ & x_2, \ x_3, \ x_4, \ x_5 \geq 0 & \end{array}$$

## Standard form and bfs

**DEFINITION OF A FEASIBLE BASIS**

$$\begin{aligned} x_1 - x_2 + x_3 &= 1 \\ -x_1 + \tfrac{1}{3}x_2 + x_4 &= 1 \\ -x_1 + \tfrac{2}{3}x_2 + x_5 &= 4 \end{aligned}$$

$$x_i \geq 0 \qquad (\, i = 1, \ldots 5 \,)$$

**System** $Ax = b, \;\; x \geq 0$

$$A = \begin{pmatrix} 1 & -1 & 1 & 0 & 0 \\ -1 & \frac{1}{3} & 0 & 1 & 0 \\ -1 & \frac{2}{3} & 0 & 0 & 1 \end{pmatrix}$$

$$B = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

**DEFINITION:** $B$ **is a feasible basis if** $\boxed{B^{-1}b \geq 0}$

$$\begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} \geq 0$$

$B$ **is a basis assocaoted to the index set {1, 4, 5}**

## Standard form and bfs

**System** $Ax = b,\ x \geq 0$

$$\begin{aligned} x_1 - x_2 + x_3 &= 1 \\ -x_1 + \tfrac{1}{3}x_2 + x_4 &= 1 \\ -x_1 + \tfrac{2}{3}x_2 + x_5 &= 4 \end{aligned}$$

$$x_i \geq 0 \qquad (\ i = 1, \ldots 5\ )$$

$B$ **basis associated to $I_B$ = {1, 4, 5}**

$$\begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} \geq 0$$

$$x_{Reord} = \begin{pmatrix} x_1 \\ x_4 \\ x_5 \\ \hline x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_B \\ \hline x_N \end{pmatrix} = \begin{pmatrix} B^{-1}b \\ \hline 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 5 \\ \hline 0 \\ 0 \end{pmatrix} \geq 0$$

## Canonical form of $Ax = b, x \geq 0$

**CANONICAL FORM OF $Ax=b$ For the index set $I_B = \{1, 4, 5\}$**

$$\begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_4 \\ x_5 \end{pmatrix} + \begin{pmatrix} -1 & 1 \\ \frac{1}{3} & 0 \\ \frac{2}{3} & 0 \end{pmatrix} \begin{pmatrix} x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix}$$

$$B \; x_B + N \; x_N = b$$

$$B^{-1}(B \; x_B + N \; x_N) = B^{-1} b$$

$$x_B + B^{-1}N \; x_N = B^{-1} b$$

$$\begin{pmatrix} x_1 \\ x_4 \\ x_5 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} -1 & 1 \\ \frac{1}{3} & 0 \\ \frac{2}{3} & 0 \end{pmatrix} \begin{pmatrix} x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix}$$

$$x_B + Y \; x_N = y_0$$

$$\begin{pmatrix} x_1 \\ x_4 \\ x_5 \end{pmatrix} + \begin{pmatrix} -1 & 1 \\ \frac{-2}{3} & 1 \\ \frac{-1}{3} & 1 \end{pmatrix} \begin{pmatrix} x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}$$

| 1 | 4 | 5 | 2 | 3 | |
|---|---|---|---|---|---|
| 1 | 0 | 0 | $-1$ | 1 | 1 |
| 0 | 1 | 0 | $-2/3$ | 1 | 2 |
| 0 | 0 | 1 | $-1/3$ | 1 | 5 |

## Canonical form of $Ax = b, x \geq 0$

**CANONICAL FORM OF A LINEAR SYSTEM $Ax=b$**

**For the set of indexes associated to a basis $B$, $I_B=\{i_1, i_2, \dots, i_m\}$**

$$x_B + Y x_N = y_0$$

$$x_B = \begin{pmatrix} x_{i_1} \\ x_{i_2} \\ \vdots \\ x_{i_m} \end{pmatrix}, \quad x_N = \begin{pmatrix} x_{j_1} \\ x_{j_2} \\ \vdots \\ x_{j_{n-m}} \end{pmatrix}$$

| $i_1$ | $\dots$ | $i_m$ | $j_1$ | $\dots$ | $j_{n-m}$ | $0$ |
|---|---|---|---|---|---|---|
| $1$ | $\dots$ | $0$ | $y_{1,1}$ | $\dots$ | $y_{1,n-m}$ | $y_{1,0}$ |
| $\vdots$ | $\ddots$ | $\vdots$ | $\vdots$ | $\ddots$ | $\vdots$ | $\vdots$ |
| $0$ | $\dots$ | $1$ | $y_{m,1}$ | $\dots$ | $y_{m,n-m}$ | $y_{m,0}$ |

Basic columns | Non-basic columns

$\geq 0$ if $B$ is a feasible basis

# 3. Geometrical aspects

## Vertexes $\longleftrightarrow$ bfs. Connection between vertices and bfs

$P = \{x \in \mathbb{R}^n \mid Ax = b,\ x \geq 0\ \}$ in standard form

- **If $\bar{x}$ is a bfs of $P$** with basic indexes $I_B \Rightarrow$ **is a vertex of $P$** :

  $\bar{x}_{Reord} = \left(\dfrac{\bar{x}_B}{0}\right) = \left(\dfrac{B^{-1}b}{0}\right)$. *Assume that it is not a vertex $P$. Then, $\exists x^1, x^2 \in F$, $x^1 \neq x^2$, $\bar{x} = \alpha x^1 + (1-\alpha)x^2$, $0 < \alpha < 1$.*
  $\left(\dfrac{\bar{x}_B}{0}\right) = \alpha \left(\dfrac{x_B^1}{x_N^1}\right) + (1-\alpha)\left(\dfrac{x_B^2}{x_N^2}\right) \Rightarrow\ x_N^1 = x_N^2 = 0$, *but, as $x^1, x^2$ are feasible:*
  $Bx_B^1 = b,\ \ Bx_B^2 = b \Rightarrow\ x_B^1 = x_B^2 = \bar{x}_B \Rightarrow\ x^1 = x^2 = \bar{x}$.

- **For each vertex $\bar{x} \in F$ exists at least a basic set $I_B$** and a basis $B$ such that: $\bar{x}_B = B^{-1}b$, $\bar{x}_N = 0$.

- **Different vertices are represented by different basis.**

## Vertexes $\longleftrightarrow$ bfs

### Existence of vertices and Fundamental Theorem in LP

Consider $P = \{x \in \mathbb{R}^n \mid Ax = b,\ x \geq 0\ \}$ in standard form.

Theorem. (2.6.5 in Bazaraa's text) If $P \neq \emptyset$, then it has at least always a vertex.

Fundamental Theorem in LP (Section 2.4 in Luenberger's text). If the feasible region of the LP is not empty, there exists always at least a basic feasible solution. Moreover, if LP has solutions, then at least one of them is a vertex.

$$\begin{array}{ll} \text{Min}_x & c^\top x \\ s.t.: & Ax = b \quad \text{(standard form)} \\ & x \geq 0 \end{array}$$

## Extreme directions

### Concept of feasible direction at $x \in S$

A vector $d$ is called a **feasible direction** at a point $x \in S$ if $x + \alpha d \in S$, for any $\alpha > 0$. ($S$ an arbitrary set)

### Definition of a Cone $C$ with vertex $0^n$

$C$ is a cone with vertex $0^n$ iff. $\forall\, x \in C$ there follows that $\mu x \in C$, $\forall\, \mu > 0$.

### Polarity

**Definition of Polar Cone**. Given an arbitrary set $S$, the polar cone of $S$ is denoted by $S^*$ (it has vertex at $0^n$) and it is defined by:

$$S^* \triangleq \{v \in \mathbb{R}^n \,|\, v^\top x \leq 0,\ \forall\, x \in S\}$$

$S^*$ is closed and convex, $S \subseteq S^{**}$; $S_1 \subseteq S_2 \Rightarrow S_2^* \subseteq S_1^*$.
If $C$ is a closed and convex cone, then $C = C^{**}$

## Extreme directions

**Positive Hull of a finite set of directions $D = \{d^1, ..., d^q\}$ (Closed, convex cone)**

The positive hull of $D$, denoted by *Pos*$\{D\}$ (also as *Cone*$\{D\}$), is defined as:

$$\textit{Pos}\{S\} \triangleq \left\{ d \in \mathbb{R}^n \mid d = \sum_{\ell=1}^{q} \mu_j d^j,\ \mu_j \geq 0,\ 1 \leq j \leq q \right\}$$

***Pos*$\{D\}$ for a general subset $S \subset \mathbb{R}^n$.**

$$d \in \textit{Pos}\{D\} \Leftrightarrow \begin{cases} \exists \{d^1, ..., d^q\} \subseteq D, q > 0, \text{ and } \mu_1, ...\mu_q \geq 0, \\ d = \sum_{\ell=1}^{k} \mu_j d^j, \end{cases}$$

As it happened with vertices (or extreme points) in a convex set, a convex cone may have directions which may not be expressed using other directions of the convex cone. These are referred to as extreme directions.

**Extreme direction $\vec{d}$ of a convex cone $D$**

$\vec{d} \in S$ is an extreme direction of $D$ iff. for any two directions $d^1 \neq d^2 \in D$ so that $\vec{d} = \alpha d^1 + (1-\alpha)d^2$, $0 \leq \alpha \leq 1$, either $\vec{d} = d^1$ or $\vec{d} = d^2$.

## Characterization of vertices and extreme directions

### Directions in a Polyhedron $P$

$d$ is a feasible direction of the polyhedron $P$ iff. $\forall\, x \in P$, $x + \mu d \in P, \forall \mu > 0$.

- Clearly, the existence of feasible directions in a Polyhedron implies its unboundedness.
- If they exist, the set of feasible directions $D$ of a Polyhedron is a **closed and convex Cone**
- If $P = \{x \in \mathbb{R}^n \mid Ax = b,\, x \geq 0\}$ (standard form), then its cone $D$ of feasible directions must verify: $Ad = 0, d \geq 0$

## Characterization of vertices and extreme directions

**(Theorem 2.6.6 in Bazaraa's text) Characterization of Extreme directions of a Polyhedron using a canonical form of $Ax = b, x \geq 0$.**

Assume that the canonical form (below) of the system $Ax = b, x \geq 0$ associated to a feasible basis $B$ is known and that columns of $A$ are reordered such that $A = (B|N)$.

$$x_B + Yx_N = Y_0, \;\; Y = B^{-1}N$$

Then, the Polyhedron $P$ has a non-null cone of feasible directions iff. for some feasible basis, the corresponding matrix $Y$ of the canonical form has at least a non-positive column $\ell$, $1 \leq \ell \leq n - m$. Furthermore, in that case, associated to the base $B$ (and to the corresponding vertex of the Polyhedron), extreme directions are of the form:

$$\vec{d} = \left( \frac{-B^{-1}a_{j_\ell}}{e_\ell} \right) = \left( \frac{-Y_\ell}{e_\ell} \right), \text{ with } Y_\ell \leq 0, \; j_\ell \in I_N$$

# Can you write the components of the extreme direction?

**CANONICAL FORM OF $Ax=b$ For the index set $I_B = \{1, 4, 5\}$**

$$\begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_4 \\ x_5 \end{pmatrix} + \begin{pmatrix} -1 & 1 \\ \frac{1}{3} & 0 \\ \frac{2}{3} & 0 \end{pmatrix} \begin{pmatrix} x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix}$$

$$B \quad x_B \quad + \quad N \quad x_N \quad = \quad b$$

$$B^{-1}(B \quad x_B \quad + \quad N \quad x_N) \quad = \quad B^{-1} b$$

$$x_B \quad + \quad B^{-1}N \quad x_N \quad = \quad B^{-1} b$$

$$\begin{pmatrix} x_1 \\ x_4 \\ x_5 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} -1 & 1 \\ \frac{1}{3} & 0 \\ \frac{2}{3} & 0 \end{pmatrix} \begin{pmatrix} x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix}$$

$$x_B \quad + \quad Y \quad x_N \quad = \quad y_0$$

$$\begin{pmatrix} x_1 \\ x_4 \\ x_5 \end{pmatrix} + \begin{pmatrix} -1 & 1 \\ \frac{-2}{3} & 1 \\ \frac{-1}{3} & 1 \end{pmatrix} \begin{pmatrix} x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}$$

| 1 | 4 | 5 | 2 | 3 | |
|---|---|---|---|---|---|
| 1 | 0 | 0 | $-1$ | 1 | 1 |
| 0 | 1 | 0 | $-2/3$ | 1 | 2 |
| 0 | 0 | 1 | $-1/3$ | 1 | 5 |

# 4. Relevant theorems

## Minkowsky-Weyl's Representation Theorem for Polyhedrons

### Theorem 2.6.7 in Bazaraa's text

Let $V = \{\hat{v}^1, ..., \hat{v}^k\}$, the set of vertexes of a polyhedron $P$, $(k > 0, i.e., V \neq \emptyset)$, and let $D = \{\vec{d}^1, ..., \vec{d}^q\}$ its (possibly empty) set of extreme directions $(q \geq 0)$.
Then $P$ can be written in the form:

$$P = \text{Hull}(V) \oplus \text{Pos}(D) \equiv \begin{cases} x \in P \Leftrightarrow x = \sum_{\ell=1}^{k} \lambda_\ell \hat{v}^\ell + \sum_{j=1}^{q} \mu_j \vec{d}^j \\ \left( \sum_{\ell=1}^{k} \lambda_\ell = 1, \ \ \lambda_\ell \geq 0, \ \ \mu_j \geq 0 \right) \end{cases}$$

## Results related to LP duality

### Farkas-Minkowsky's lemma. Theorem 2.4.5 in Bazaraa's text

Let $A \in \mathbb{R}^{m \times n}$ 1) and $b \in \mathbb{R}^m$. Then, 1) is equivalent to 2)

1. $P = \{x \in \mathbb{R}^n \mid Ax = b,\ x \geq 0\} \neq \emptyset$
2. $\forall u \in \{v \in \mathbb{R}^m \mid A^\top v \geq 0\}$, it is verified that $b^\top u \geq 0$

### Gale's theorem

Either 1) or 2) holds but not both:

1. $A^\top u \leq c$ has a solution $u$.
2. $Ad = 0$, $c^\top d = -1$, $d \geq 0$ has a solution $d$.

# 5. A worked example

5. Calculate the basic feasible solutions for the set of constraints:

$$\begin{aligned} x_1 - x_2 + x_3 &= 1 \\ -x_1 + \tfrac{1}{3}x_2 + x_4 &= 1 \\ -x_1 + \tfrac{2}{3}x_2 + x_5 &= 4 \\ x_i \geq 0 & \qquad (\, i = 1, \ldots 5 \,) \end{aligned}$$

starting from the basis $I_B = \{\, 3, 4, 5 \,\}$.

| 3 | 4 | 5 | 1 | 2 | |
|---|---|---|---|---|---|
| 1 | 0 | 0 | 1 | −1 | 1 |
| 0 | 1 | 0 | −1 | 1/3 | 1 |
| 0 | 0 | 1 | −1 | 2/3 | 4 |

$$I_2^+ = \{i_4 = 2, \ i_5 = 3\}$$

$$\hat{\varepsilon} = Min_{\ i \in I_2^+} \ \{ \frac{y_{i,0}}{y_{i,2}} \} = Min\{ \frac{1}{1/3} , \frac{4}{2/3} \} = \frac{1}{1/3} = 3 \Rightarrow i_p = 2 \Rightarrow p = 4$$

exiting variable $x_4$.

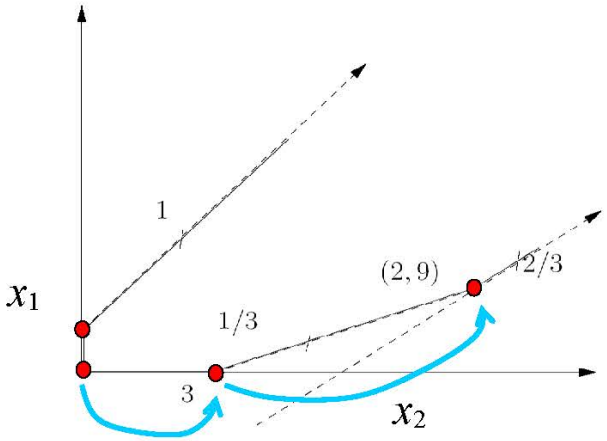
$$\begin{array}{ccccc|c} 3 & 4 & 5 & 1 & 2 & \\ \hline 1 & 0 & 0 & 1 & -1 & 1 \\ 0 & 1 & 0 & -1 & 1/3 & 1 \\ 0 & 0 & 1 & -1 & 2/3 & 4 \end{array} \rightarrow \begin{array}{ccccc|c} 3 & 4 & 5 & 1 & 2 & \\ \hline 1 & 0 & 0 & 1 & -1 & 1 \\ 0 & 3 & 0 & -3 & 1 & 3 \\ 0 & 0 & 1 & -1 & 2/3 & 4 \end{array}$$

$$\begin{array}{ccccc|c} 3 & 4 & 5 & 1 & 2 & \\ \hline 1 & 3 & 0 & -2 & 0 & 4 \\ 0 & 3 & 0 & -3 & 1 & 3 \\ 0 & -2 & 1 & 1 & 0 & 2 \end{array}$$

New basis $I_B = \{3, 2, 5\}$

$$I_1^+ = \{i_5 = 3\}$$

$$\hat{\varepsilon} = Min_{\ i \in I_1^+} \ \{ \frac{y_{i,0}}{y_{i,2}} \} = Min\{ \frac{2}{1} \} = 2 \Rightarrow$$




$$I_B = \{3, 2, 1\}$$

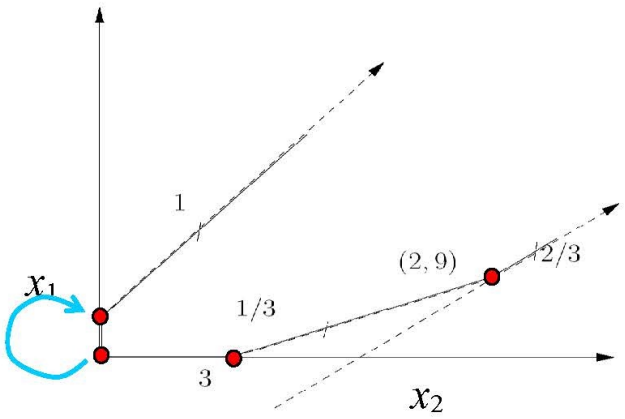
| 3 | 4 | 5 | 1 | 2 | |
|---|---|---|---|---|---|
| 1 | −1 | 2 | 0 | 0 | 8 |
| 0 | −3 | 3 | 0 | 1 | 9 |
| 0 | −2 | 1 | 1 | 0 | 2 |

$$d^\top = (d_3, d_2, d_1, d_4, d_5) = (1, 3, 2, 1, 0)$$

# Could you apply Minkowsky-Weyl's theorem ?

| 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|
| 1 | $-1$ | 1 | 0 | 0 | 1 |
| $-1$ | $1/3$ | 0 | 1 | 0 | 1 |
| $-1$ | $2/3$ | 0 | 0 | 1 | 4 |

Enters $x_1 \Rightarrow$ Exits $x_3 \Rightarrow$

| 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|
| 1 | $-1$ | 1 | 0 | 0 | 1 |
| 0 | $-2/3$ | 1 | 1 | 0 | 2 |
| 0 | $-1/3$ | 1 | 0 | 1 | 5 |

$$d^\top = (d_1, d_4, d_5, d_2, d_3) = (1, 2/3, 1/3, 1, 0)$$